

Duc Hoang

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SUMMARY

High-energy physicist specializing in machine learning, data science, and real-time FPGA systems. Extensive experience delivering production-level ultra-low-latency ML inference on FPGAs and petabyte-scale data analysis for CERN's CMS experiment. Proven leader in multi-institutional projects, skilled in developing custom ML solutions and deploying robust, high-performance software.

EDUCATION

- **Massachusetts Institute of Technology (MIT)** 2021 - 2026
PhD, High Energy Physics. Thesis: Tagging Jets at the data frontier Cambridge, USA
- **Rhodes College** 2017-2021
B.S., Physics, Summa Cum Laude. Thesis: Triggering on electron-nucleon scattering events at LDMX Memphis, USA

PROFESSIONAL EXPERIENCE

- **MIT & European Organization for Nuclear Research (CERN)** 2021 - Present
Graduate Researcher & Next Generation Trigger PhD Fellow Cambridge, USA & Geneva, Switzerland
 - Coordinated multi-institutional team to develop **ultra-low-latency FPGA-accelerated ML models** for CMS Level-1 Trigger, which processes approximately 10% of global internet traffic in real time. [Hear me explain it along the trigger leadership team](#) [1]
 - Built first **FPGA-based ML algorithms for real-time jet tagging** and delivered cross-platform, low-latency data transfer tests between different FPGA board platforms. [2]
 - Analyzed all **Run 2 CMS data (a few PBs)** to deliver **most sensitive Higgs $\rightarrow b\bar{b}$ measurement in the V(qq)H(bb) decay channel**, invited presentation at Rencontres de Moriond 2025. [3]
 - Designed and led practical CERN training sessions on machine **learning techniques for FPGA-based Level-1 Trigger applications** for CERN students and researchers.
 - Created and taught data analysis projects for **MIT 8.16/8.316: Data Science in Physics**, which is published on MITx course [Computational Data Science in Physics](#).
 - Supervised MIT undergraduates on research projects involving the CMS Level-1 Trigger system and the development of **novel machine learning architectures on FPGAs**. [8].
- **Fermi National Accelerator Laboratory** 2019 - 2021
Undergraduate Researcher Chicago, USA
 - Developer of [hls4ml](#), A package for **ultra-low-latency machine learning inference in FPGAs** using high level synthesis language (HLS). [4]
 - Designed superconducting magnet quench detection systems [5]
 - Analyzed simulation data to predict potential signal gains for the real-time trigger systems for Light Dark Matter Experiment (LDMX). [6]
 - Machine Learning analysis to infer hyper-parameter optimization performance of models used in the MINERvA Experiment. [7]

SELECTED PUBLICATIONS & SOFTWARE

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| [1] | CMS Collaboration , "The Phase-2 Upgrade of the CMS Level-1 Trigger", https://cds.cern.ch/record/2714892 | superconducting magnet quenches," in <i>IEEE Transactions on Applied Superconductivity</i> , doi: 10.1109/TASC.2021.3058229 . |
| [2] | CMS Collaboration , 2025. "Multiclass object tagging in the CMS Phase-2 Level-1 Correlator Trigger." CMS Detector Performance Summary. https://cds.cern.ch/record/2936315 . | [6] LDMX Collaboration , "Current Status and Future Prospects for the Light Dark Matter eXperiment" [arXiv:2203.08192][hep-ex]. |
| [3] | CMS Collaboration , "Search for a boosted Higgs boson decaying to bottom quark pairs in association with a hadronically decaying W or Z boson with the CMS detector using proton-proton collisions at $\sqrt{s} = 13$ TeV", https://cds.cern.ch/record/2928061 | [7] D. Hoang et al. "Inferring Convolutional Neural Networks' Accuracies from Their Architectural Characterizations," <i>2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA)</i> , Boca Raton, FL, USA, 2019, pp. 1388-1391. DOI: 10.1109/ICMLA.2019.00226 |
| [4] | hls4ml Community , " hls4ml: An Open-Source Codesign Workflow to Empower Scientific Low-Power Machine Learning Devices " | [8] CMS Collaboration . 2024. "NNPuppiTaus: PUPPI Tau Reconstruction in the Level-1 Trigger with Real-Time Machine Learning." CMS Detector Performance Summary. https://cds.cern.ch/record/2895660 . |
| [5] | D. Hoang et al. , "Intelliquench: an adaptive machine learning system for detection of | |

SKILLS

- **Specialized Area:** Big Data Analysis, Statistical Modeling, ultra-low latency FPGA Inference, Machine Learning
- **Programming Languages (years of experience):** Python (7), C++ (5), HTML(1), CSS(1), TypeScript(1)
- **Web Technologies:** Django, Angular
- **Database Systems:** SQLite, PostgreSQL
- **Data Science & Machine Learning:** Most Python machine learning and data science tools.
- **Cloud Technologies:** Google Cloud
- **DevOps & Version Control:** Continuous Integration, Docker, Singularity / Apptainer, Git

HONORS AND AWARDS

- **The 2025 Breakthrough Prize in Fundamental Physics** 2025
Received as a member of the CMS Collaboration 
 - The prize is awarded to physicists from theoretical, mathematical, or experimental physics that have made transformative contributions to fundamental physics, and specifically for recent advances.
- **2025 European Physical Society Conference on High Energy Physics (EPS-HEP) Poster Prize** 2025
European Physical Society (EPS) 
 - Awarded for best poster at EPS-HEP 2025 (out of more than 100 posters), a highly prestigious biannual international conference featuring major results from leading high-energy physics experiments and collaborations.
- **MIT Public Service Fellow (PKG) Fellowship** 2025
Priscilla King Gray Public Service Center - MIT 
 - Founded and organized, along with other International Olympiad Medalists from Central Vietnam, a STEM [summer school](#), providing under-privileged high school students in Central Vietnam the opportunity to study emerging fields such as artificial intelligence, data science, programming, renewable energy, and quantum technology at no cost.
- **Next Generation Trigger PhD fellow.** 2024
CERN 
 - Awarded for exceptional PhD students to work along CERN scientists to develop the next generation trigger system at the Large Hadron Collider.
- **2024-25 Christiaan Huygens Graduate Fellowship - MIT.** 2024
MIT School of Science
 - Internal Fellowship providing financial support for the full academic year in the School of Science for students with non-traditional backgrounds.
- **MIT-The Royal Game of Ur Champion** 2023
MIT Playur II Tournament 
 - The Royal Game of Ur is the oldest board game in the world.
- **Henry W Kendall (1955) first year graduate fellowship.** 2021
MIT Laboratory for Nuclear Science
 - Financial support for first year graduate students.
- **Miscellaneous Undergraduate Awards.** 2017-2021
Rhodes College
 - Research Award in Physics, Council on Undergraduate Research (CUR) Physics & Astronomy Student Travel Award, W.O Shewmaker Award for Excellence in First-Year Search Program, Award for Excellence in Physics by a First-Year Student, Summa Cum Laude, Sigma Pi Sigma, Phi Beta Kappa, Physics Honors.